



CARIBBEAN EXAMINATIONS COUNCIL

ADVANCED PROFICIENCY EXAMINATION

PHYSICS

UNIT 2 – Paper 02

2 hours 30 minutes

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C.E.S.E.C.

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of **SIX** questions, divided into Sections A and B.
2. Section A consists of **THREE** questions. Candidates must attempt **ALL** questions in this section. Answers for this section must be written in the spaces provided in this question paper.
3. Section B consists of **THREE** questions. Candidates must attempt **ALL** questions in this section. Answers for this section must be written in the spaces provided in this question paper.
4. Diagrams may be drawn in pencil but **all writing** must be in **ink**.
5. All working **MUST** be **CLEARLY** shown.
6. The use of silent non-programmable calculators is permitted, but candidates should note that the use of an inappropriate number of figures in answers will be penalised.

LIST OF PHYSICAL CONSTANTS

Speed of light in free space	c	=	$3.00 \times 10^8 \text{ m s}^{-1}$
Permeability of free space	μ_0	=	$4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity of free space	ϵ_0	=	$8.85 \times 10^{-12} \text{ F m}^{-1}$
	$\frac{1}{4\pi\epsilon_0}$	=	$9.0 \times 10^9 \text{ m F}^{-1}$
Elementary charge	e	=	$1.60 \times 10^{-19} \text{ C}$
Planck's constant	h	=	$6.63 \times 10^{-34} \text{ J s}$
Unified atomic mass constant	u	=	$1.66 \times 10^{-27} \text{ kg}$
Rest mass of electron	m_e	=	$9.11 \times 10^{-31} \text{ kg}$
Rest mass of proton	m_p	=	$1.67 \times 10^{-27} \text{ kg}$
Acceleration due to gravity	g	=	9.81 m s^{-2}
1 Atmosphere	Atm	=	$1.00 \times 10^5 \text{ N m}^{-2}$
Avogadro's constant	N_A	=	$6.02 \times 10^{23} \text{ per mole}$

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SECTION A

Answer ALL questions.

You MUST write your answers in the spaces provided in this answer booklet.

1. (a) Define magnetic flux density, B , in terms of the force on a current-carrying wire.

[2 marks]

- (b) Figure 1 (b) shows a wire, P, carrying a current perpendicular to the plane of the paper, between two flat permanent magnets which have poles on their faces as shown in Figure 1 (a). Draw the resultant magnetic field pattern between the two magnets. Indicate on the figure, with an arrow labelled M, the direction of motion of the wire.

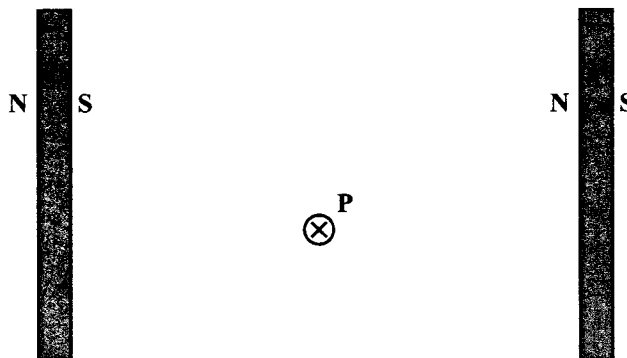


Figure 1 (b)

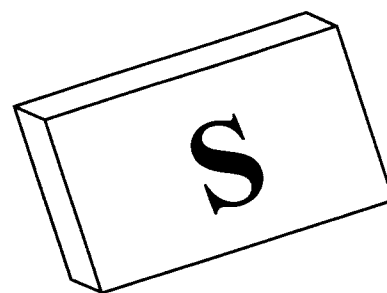


Figure 1 (a)

[4 marks]

- (c) Figure 1 (c) shows a narrow beam of electrons travelling at a speed, v , and directed into a uniform electric field between two oppositely charged parallel plates placed a distance, d , apart. The top plate is at a positive potential, V , relative to the lower plate. A magnetic field is now applied perpendicularly to the direction of the electric field between the plates and the deflection of the beam is cancelled.

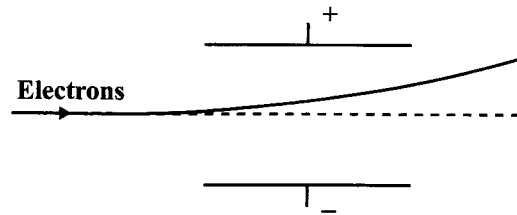


Figure 1 (c)

- (i) State the direction of the magnetic field.

[1 mark]

- (ii) By considering the forces on EACH electron, show that the magnetic flux density is given by

$$B = \frac{V}{vd} \quad .$$

[4 marks]

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- (iii) Given that $V = 3500 \text{ V}$, $v = 2.8 \times 10^7 \text{ m s}^{-1}$, and $d = 50 \text{ mm}$, calculate the magnetic flux density of the magnetic field.

[2 marks]

- (iv) The magnetic flux density is now doubled. The distance between the plates is then adjusted so that no deflection occurs. Calculate the new distance.

[2 marks]

Total 15 marks

2. (a) (i) What does the term 'bistable' mean when used in reference to an electronic circuit?

[1 mark]

- (ii) Figure 2 (a) shows a quad NOR circuit board with l.e.d.'s to show the output of each gate. It is connected to a 6 V battery.

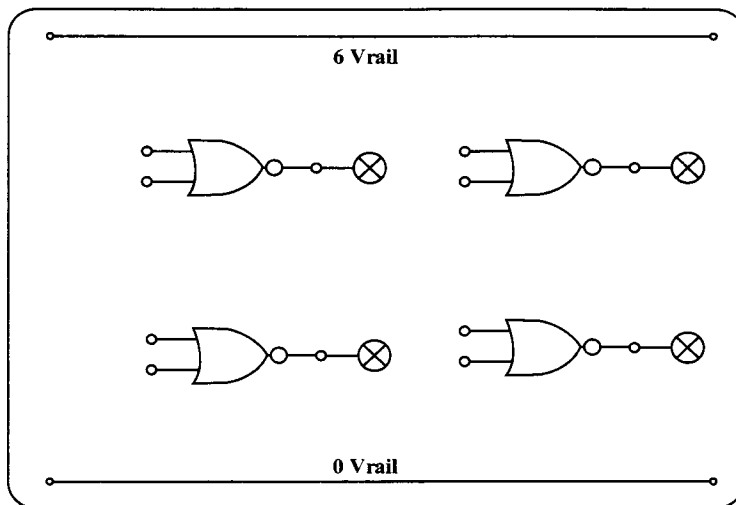


Figure 2 (a)

Draw connecting wires on the diagram to show the construction of an S-R flip-flop with clearly marked inputs I_1 and I_2 and outputs Q and $\bar{Q}$.

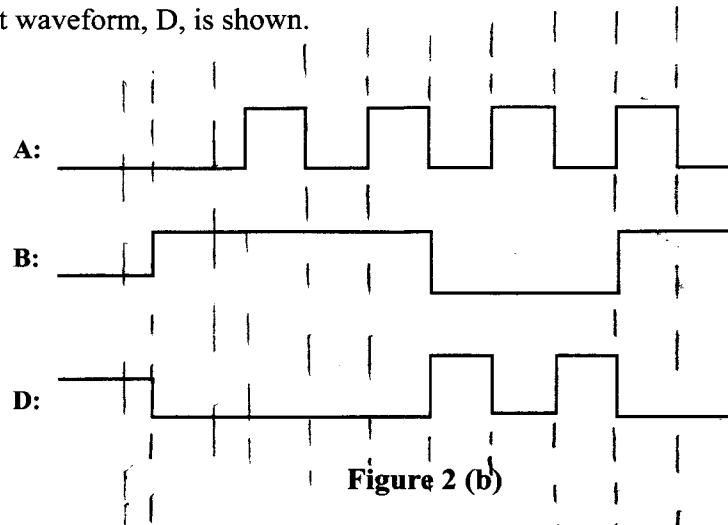
A friend asks you to demonstrate how this circuit acts as an electronic latch, perhaps as part of a burglar alarm. Complete the sequential truth table on the right and use it to assist you in describing the connections you would need to make to show this feature to your friend.

Row No.	sequence of inputs		outputs	
	I_1	I_2	Q	$\bar{Q}$
1	0	1		
2	0	0		
3	1	0		
4	0	0		
5	1	0		
6	0	0		

[5 marks]

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- (b) Figure 2 (b) shows two wave forms A and B as inputs to a logic circuit. The resulting output waveform, D, is shown.



Draw the truth table for the circuit and hence state the equivalent logic gate for the circuit.

Truth table:

Equivalent logic gate:

[4 marks]

- (c) A 1 kHz square wave of amplitude 5 V is input into one input of an XOR gate while the other input is grounded. An oscilloscope is used to view the output waveform.

(i) What is the frequency of the XOR output?

[1 mark]

(ii) If the same frequency is input but the other input to the XOR is tied to 5 V, what is the frequency of the output?

[1 mark]

(iii) The same square wave is applied to one input of logic gate P while the other input is grounded (See Figure 2 (c) below). The output waveform is a flat line at 0 V. State the name of logic gate P and use a truth table to explain your answer.

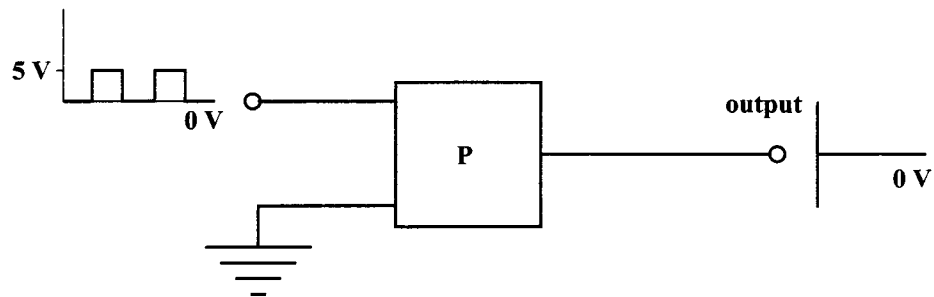


Figure 2 (c)

[3 marks]

Total 15 marks

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3. (a) In Figure 3 (a), the arrows represent the types of radiation emitted by a radioactive source. The radiation is stopped by the materials shown. Identify EACH type of radiation by writing the name of the radiation next to the arrow that represents the type of radiation.

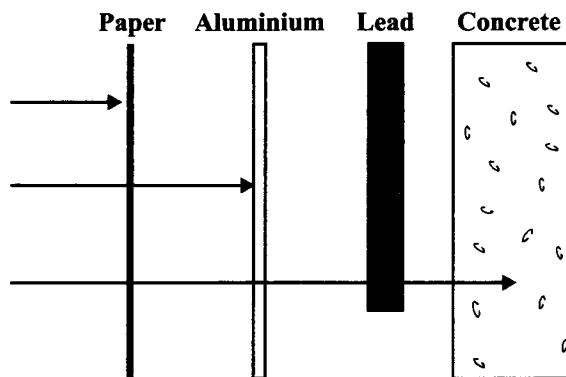


Figure 3 (a)

[3 marks]

- (b) Figure 3 (b) shows an arrangement for investigating the variation of intensity of gamma (γ) rays with distance from the small source, S. The distance, d , is taken from two convenient reference points as shown and should be reduced by an amount, x , to give the correct distance. The background count of the tube was 27 counts per minute. The count rate is a direct measure of the intensity of the gamma rays.

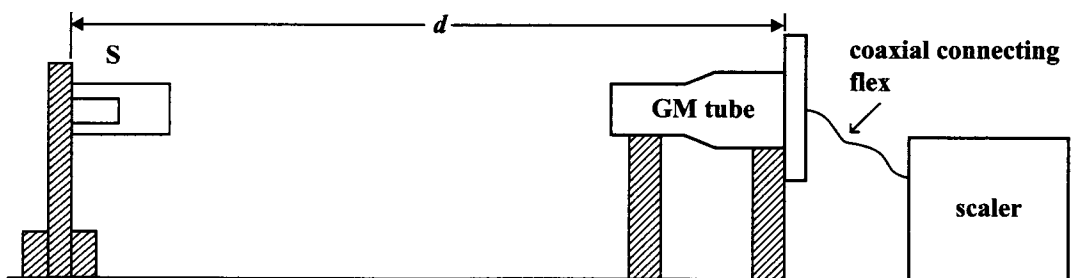


Figure 3 (b)

The following results were obtained and are shown in Table 1.

TABLE 1

d/cm	Count Rate (Counts/min)	Corrected Count Rate (C/min)	$\frac{I}{\sqrt{C}}$
15	2527		
20	1135		
25	652		
30	427		
35	301		
40	235		

The intensity, C , of the γ rays is thought to be inversely proportional to the square of the corrected distance, $(d - x)$.

- (i) Express this relationship mathematically.

[1 mark]

- (ii) Show that a straight line graph would be expected if d were plotted against $\frac{I}{\sqrt{C}}$.

[2 marks]

- (iii) Complete Table 1 to show the data that must be plotted to obtain the graph.

[2 marks]

- (iv) On the graph paper provided on page 11, plot the graph.

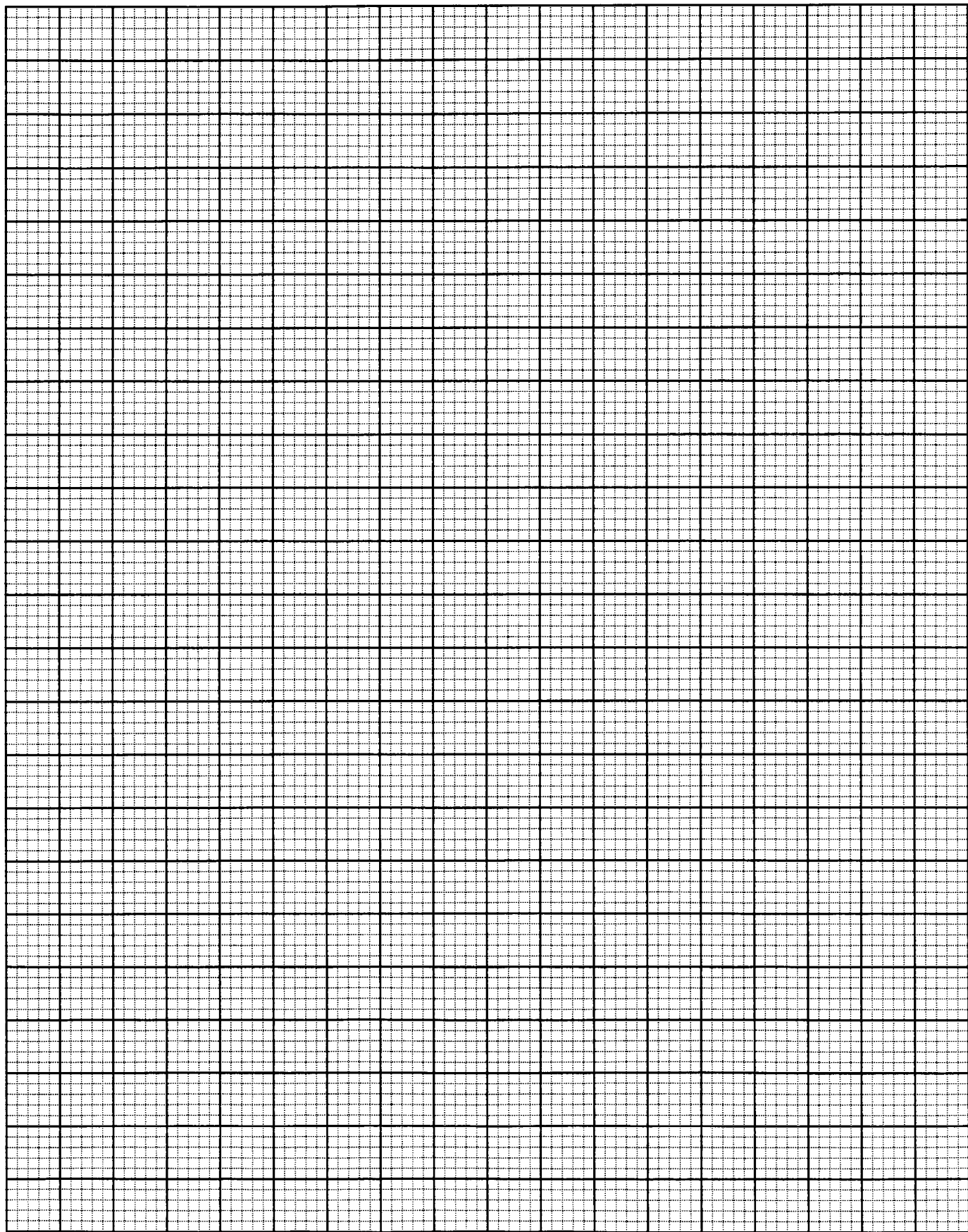
[5 marks]

- (v) Determine, from the graph, the value of x .

[2 marks]

Total 15 marks

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SECTION B

Answer ALL questions.

Write your answers in the spaces provided at the end of the questions.

4. (a) Derive an equation for the equivalent capacitance of **two** capacitors connected in series. [3 marks]
- (b) A “black box” contains a $4.7 \mu\text{F}$ capacitor in series with another capacitor with unknown capacitance, C . To collect the data for the graph below [Figure 4] it was connected to a battery with emf, E and then discharged through a $10 \text{ M}\Omega$ resistor. Describe, with the aid of a circuit diagram, how the experiment should be performed, assuming any other required apparatus is available in the laboratory. [4 marks]
- (c) Figure 4 shows a graph depicting the experimental results obtained using the black box for the discharge of the capacitors through the resistor. The decay voltage, V , varies with time, t , according to the equation

$$V = V_0 \exp\left(\frac{-t}{RC_{\text{eq}}}\right), \text{ where } C_{\text{eq}} \text{ is the capacitance of the two capacitors in series.}$$

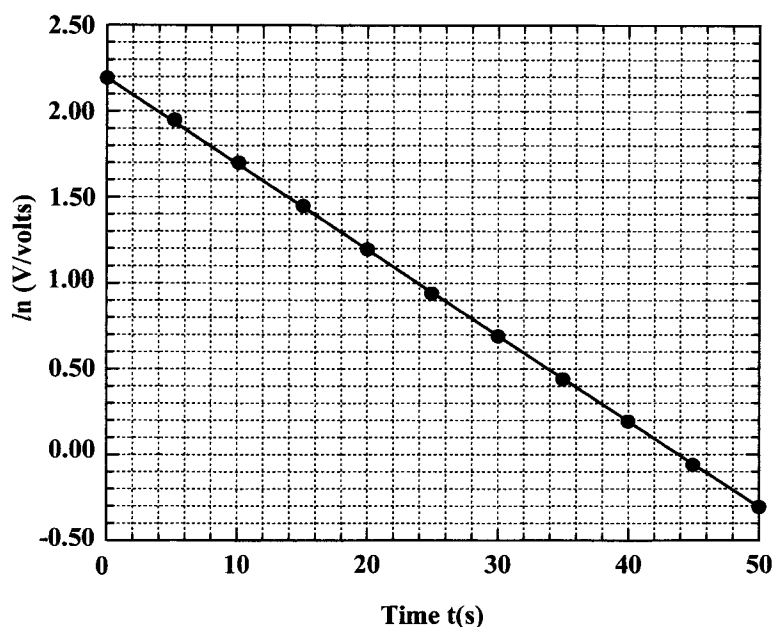


Figure 4

- (i) Calculate the gradient of the graph and hence determine the time constant of the circuit. [4 marks]
- (ii) Calculate the value of the unknown capacitance, C . [3 marks]
- (iii) Determine the value of the unknown source of emf, E . [1 mark]

Total 15 marks

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You MUST write the answer to Question 4 here.

4. (a)

(b)

(b)

(c)

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5. (a) (i) Explain the terms 'bandwidth' and 'feedback' as applied to operational amplifiers (op-amps). [2 marks]

- (ii) Figure 5 (a) shows an op-amp on open loop. An alternating sinusoidal voltage of peak-to-peak voltage 2.0 V is applied to the inverting input, with the non-inverting input terminal, Q, connected to earth. The circuit saturates at ± 5 V.

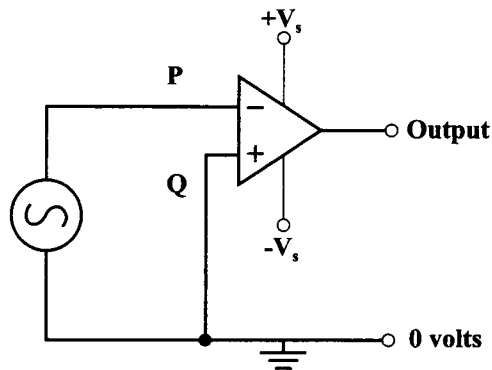


Figure 5 (a)

Sketch, on the same axes, the input waveform and the output waveform produced from this circuit. [3 marks]

- (iii) The non-inverting input, Q, is then connected to a potential divider and a constant voltage of 0.5 V is applied to Q. Figure 5 (b) shows the output waveform produced by this arrangement.

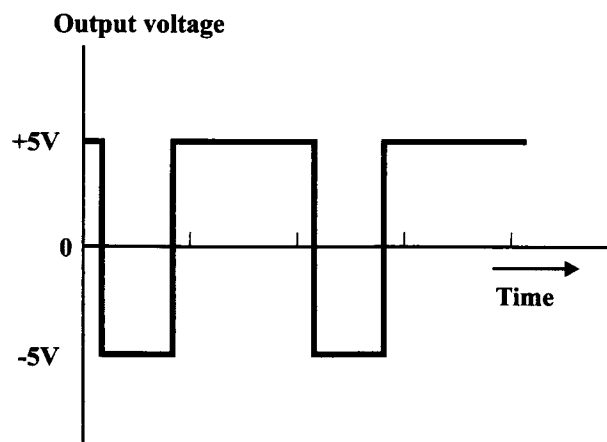


Figure 5 (b)

Explain the shape of this waveform.

[2 marks]

- (b) The circuit in Figure 5 (a) is modified by adding a $2.0\text{ M}\Omega$ feedback resistor and a $0.4\text{ M}\Omega$ resistor between the inverting input, P, and earth. The a.c. supply unit is disconnected from P and reconnected between Q and earth.

(i) Draw the resulting circuit and state the name of this type of circuit. [3 marks]

(ii) Calculate the voltage gain of the circuit. [3 marks]

(iii) Calculate the MAXIMUM input voltage for no saturation at the output. [2 marks]

Total 15 marks

You MUST write the answer to Question 5 here.

5. (a) (i) _____

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(ii)

(iii)

[illegible]

(b) (i)

(ii)

(iii)

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6. (a) (i) Explain what is meant by 'nuclear fission'.
- (ii) Sketch a graph to show how the binding energy per nucleon varies with the mass number of the nucleus. Show on the graph the approximate positions of carbon-12 and uranium-235.
- (iii) Use your graph to explain why energy is released when a heavy nucleus undergoes fission.

[6 marks]

- (b) The fission of 1 kg of uranium-235 releases as much thermal energy as 50 000 kg of oil. 30 MJ of energy is released when 1 kg of oil is burned.

Calculate the mass of fuel used per day, in EACH case, in a 1 000 MW power station which is only 25% efficient and which uses

- (i) oil
- (ii) uranium.

[8 marks]

- (c) Hence comment on the sustainability of using oil as an energy source for the next century.

[1 mark]

Total 15 marks

You MUST write the answer to Question 6 here.

6. (a)

(b)

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(b)

(c)

END OF TEST